Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.DOI

MM-Net: Joint Sleep Staging, Respiratory-Event Detection and Apnea–Hypopnea Estimation After Ischemic Stroke With a Fine-Tuned EEG Foundation Encoder, Validated on Five Cohorts

MD WAHIDUZZAMAN SUVA[1], ESM-E MOULA CHOWDHURY ABHA[1], MD IMTIAJ ALAM SAJIN[1], AND MD IFTEKHARUL MOBIN[1]

[1]Department of Computer Science and Engineering, American International University-Bangladesh, Dhaka 1229, Bangladesh

Corresponding author: Md Iftekharul Mobin (e-mail: iftekhar.mobin@aiub.edu).

The authors received no specific funding for this work.

ABSTRACT Sleep-disordered breathing affects most patients after ischemic stroke and predicts poorer recovery, yet automated sleep analysis reads the electroencephalogram and reports stages alone. We present MM-Net, a multimodal multi-task network that reads the full polysomnogram and returns, in one pass, the sleep stage of every 30-second epoch, a per-epoch respiratory-event score and the patient's apnea–hypopnea index (AHI). Its neural stream carries an EEG foundation encoder (LaBraM) fine-tuned on sleep with both EEG and ocular channels, and we show that such an encoder improves the downstream sequence model only when its training embeddings are cross-fitted and ridge-aligned to the deployed encoder; we introduce that procedure. On iSLEEPS (99 stroke patients, ten-fold patient-independent cross-validation) MM-Net stages sleep at 0.763 accuracy (Cohen's κ 0.656), ahead of all eight deep baselines retrained on the same folds in every fold (by 0.042 to 0.164; Holm $p = 0.016$), detects respiratory events at 0.825 AUC against 0.673 for gradient boosting, and estimates the AHI with Spearman $\rho = 0.765$. The gains generalize. Trained and tested on the 100-patient ISRUC-I cohort, MM-Net reaches 0.803 accuracy (κ 0.744) and respiratory AUC 0.889, and it reaches 0.864 on healthy sleep (Sleep-EDF). Trained on iSLEEPS alone and applied unchanged, it stages the 100 ISRUC-I patients at 0.730 and, on ISRUC-I and a 25-patient apnea cohort, detects respiratory events at 0.83–0.84 AUC with AHI ρ of 0.63–0.65. MM-Net is, to our knowledge, the first model to deliver staging, respiratory-event detection and AHI jointly for stroke polysomnography. Code and features are released.

INDEX TERMS Apnea–hypopnea index, EEG foundation model, ischemic stroke, multi-task learning, multimodal learning, polysomnography, sleep apnea, sleep staging.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3 and rapid-eye-movement (REM) sleep, following the American Academy of Sleep Medicine (AASM) rules [1]. Automated scoring has progressed from convolutional and recurrent encoders [2]–[4] through attention and sequence models [5]–[7] to large self-supervised EEG foundation models [8]–[11], and on healthy sleepers these methods reach about 85% accuracy [12], [13].

The patients who need sleep analysis most are not healthy sleepers. After an ischemic stroke, sleep-disordered breathing is common and is tied to worse functional recovery [14], [15], and the lesion itself distorts the spindles, slow waves and muscle atonia that staging relies on [16]–[18]. A clinical polysomnogram (PSG) records everything needed to address both problems at once, the EEG, ocular and muscle activity, the electrocardiogram (ECG), airflow, thoracic and abdominal effort and oxygen saturation (SpO_2), yet automated tools read the EEG and report the stage, leaving the breathing disorder to manual scoring. On iSLEEPS [19], the first public stroke polysomnography corpus, the median patient carries a scored respiratory event in 13% of epochs (Fig. 1), and architectures built on healthy sleep lose 12 to 20 accuracy points [20].

We present MM-Net, a model that reads the whole polysomnogram of a stroke patient and returns the three quantities a sleep clinician reports: the hypnogram, the respiratory events and the apnea–hypopnea index (AHI). Its neural stream carries a pretrained EEG foundation encoder, LaBraM [8], fine-tuned on sleep. Using a fine-tuned encoder inside a second model trained on the same patients raises a problem that frozen encoders do not have: the encoder has seen its own training patients, so the embeddings the downstream model learns from are in-sample while those it is tested on are not. MM-Net addresses this by *cross-fitting* the training embeddings and *aligning* each cross-fitted encoder to the deployed one.

Our contributions are:

1) **Cross-fitted, aligned foundation-encoder integration.** We fine-tune LaBraM on EEG and ocular channels and integrate it into a multimodal sequence model through cross-fitting and ridge alignment of the embedding spaces (Section IV-C), and isolate the contribution of each step.
2) **A single model for staging, respiratory events and the AHI.** MM-Net fuses the foundation-encoder representation with engineered neural descriptors and a learned encoder of eight raw cardiorespiratory channels, and derives a calibrated AHI estimate from its respiratory head over its own decoded sleep (Section IV).
3) **A same-fold benchmark on stroke sleep.** We compare MM-Net with reimplemented deep staging architectures and conventional respiratory detectors on identical patient-independent folds of the iSLEEPS stroke cohort (Section VI).
4) **Validation across five cohorts and three settings.** We evaluate MM-Net in-cohort, trained and tested on the public ISRUC-Sleep and Sleep-EDF cohorts, and applied without retraining to sleep-disorder, healthy and apnea cohorts recorded on other systems (Section VII).
5) **Physiological attribution and deployment guidance.** We attribute each output to the physiology it depends on by interventional ablation, test whether the respiratory read-out depends on the channel the events are scored from, and identify which inputs transfer across acquisition systems (Sections VI-F and VII).

II. RELATED WORK

A. DEEP SLEEP STAGING

DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep added attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4], [7]; XSleepNet fused raw and spectral views [6]; SleepTransformer added interpretability and uncertainty [21]; L-SeqSleepNet models whole-night cycles [22]; and recent backbones explore flexible channel sets and state-space models [23]–[26]. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [27]. These models set accuracy on healthy corpora but depend on the cortical microevents that stroke disturbs, and they transfer poorly to stroke patients [20].

B. EEG FOUNDATION MODELS

LaBraM, CBraMod, EEGPT and BIOT learn transferable EEG representations from large unlabeled corpora [8]–[11]; SleepFM extends the idea to several PSG modalities [28], and cross-scale models and benchmarks continue to appear [29]–[32]. These encoders are usually either frozen or fine-tuned end to end on a single task. How to place a *fine-tuned* encoder inside a second, downstream model trained on the same patients has received little attention, although it is the standard route to combining a foundation representation with task-specific signals. The problem is the one stacked generalization addresses [33]: a second-stage model must be trained on first-stage outputs that are out of sample, which cross-fitting provides [34]. We add the step this setting needs and stacking does not, aligning the embedding spaces of the cross-fitted encoders to the deployed encoder.

C. SLEEP-DISORDERED BREATHING AND JOINT MODELS

Apnea and hypopnea detectors read airflow and effort shape, oximetry and ECG-derived heart-rate variability, usually at fine time resolution and for that task alone [1]. Huttunen *et al.* trained one network to score sleep stages and respiratory events simultaneously on 877 clinical recordings [35], and multitask staging-and-apnea models have followed in elderly populations [36]. None has been shown on stroke, where the neural signal is itself damaged, and none combines a foundation encoder with a cardiorespiratory stream or reports a validated AHI estimate across cohorts.

D. SLEEP AFTER STROKE

Sleep-disordered breathing is common in the subacute phase and linked to poorer outcome [14], [15], [37], and focal injury changes spindles, slow waves and REM atonia [16]–[18], [38]–[40]. Automated staging on ischemic stroke has so far been reported only as single-task baselines on iSLEEPS [19], [20]. MM-Net addresses staging, respiratory events and the AHI jointly on this population and validates each output beyond it.

III. COHORTS AND PREPROCESSING

a: iSLEEPS

The corpus holds overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male), scored into the five AASM stages [19]. The records of SN28 are byte-identical to those of SN15, so SN28 is excluded, giving $N = 99$ and 92,560 epochs (Table 2). The neural stream reads four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), two EOG channels (E1:M2, E2:M2) and the submental EMG at 100 Hz. The cardiorespiratory stream reads ECG, oronasal thermal airflow, nasal pressure, thoracic, abdominal and summed effort, SpO_2 and pulse, resampled by polyphase filtering to 25 Hz. Snore is excluded because respiratory events are

TABLE 1. The five cohorts. iSLEEPS is the development cohort; the others test generalization in two settings: trained and tested on the cohort with patient-independent folds (CV), and trained on iSLEEPS and applied unchanged (transfer).

Cohort	Population	Patients	Nights	Signals used	Respiratory events	Use
iSLEEPS [19]	subacute ischemic stroke	99	99	full PSG, 8 cardioresp. ch.	scored	development, 10-fold CV
ISRUC-I [41]	sleep disorders	100	100	full PSG, 8 cardioresp. ch.	scored	10-fold CV; transfer
ISRUC-II [41]	sleep disorders	8	16	full PSG, 8 cardioresp. ch.	scored	CV (with III); transfer
ISRUC-III [41]	healthy	10	10	full PSG, 8 cardioresp. ch.	scored	CV (with II); transfer
UCD [42], [43]	suspected sleep apnea	25	25	C3/C4 EEG, 7 cardioresp. ch.	scored	transfer
Sleep-EDF [44]	healthy	21	40	2 EEG + EOG	none	10-fold CV

scored partly from the same airway signal; limb, position and housekeeping traces carry no staging or breathing information.

b: Epoch alignment

iSLEEPS retains only epochs scored as a sleep stage, so a recording whose hypnogram contains unscored rows has fewer retained epochs than 30-second slots on the clock. We cut every cardiorespiratory signal and every respiratory-event label at the true recording time of each retained epoch. Cutting them on the clock grid instead silently shifts them against the EEG after the first dropped row; in iSLEEPS this affects 63 of the 100 released recordings, by up to 137 epochs. Respiratory events are mapped to epochs by onset, and the annotated AHI is the number of apneas and hypopneas starting in scored sleep per hour of scored sleep.

c: External cohorts

ISRUC-Sleep [41] uses the same central and occipital derivations against the contralateral mastoid and records SpO_2, airflow, nasal pressure, thoracic and abdominal effort and ECG; summed effort is formed as thorax plus abdomen, and pulse as beat-to-beat heart rate from ECG R-peaks. Subgroup I comprises 100 patients with sleep disorders, II eight patients recorded on two nights, and III ten healthy adults; annotations of the first rater are used. In 30 of the 126 ISRUC recordings the EDF header declares the EEG range as $\pm 25\,\mu$V, which places N3 at 9–23 μV peak to peak against 98–283 μV in the other recordings and below the 75 μV that defines N3; we restore these channels with a factor of ten, a correction decided from the header alone. The UCD database [42] holds 25 adults referred for suspected sleep-disordered breathing, with AHI from 1.2 to 81.2; it records only C3 and C4, which also feed the occipital inputs, and its headers carry no physical unit, so EEG, EOG and EMG are scaled per recording to the iSLEEPS median whole-night amplitude, with no label used. Sleep-EDF Cassette [44] provides 40 healthy nights from 21 subjects, with Fpz-Cz and Pz-Oz mapped to the central and occipital inputs.

IV. MM-NET

A. FORMULATION

A night is a sequence of T epochs. For epoch t MM-Net receives a neural vector $\mathbf{f}_t = [\mathbf{d}_t; \mathbf{e}_t] \in \mathbb{R}^{388}$, the concatenation of 188 engineered neural descriptors $\mathbf{d}_t$ and a 200-dimensional foundation embedding $\mathbf{e}_t$, and a raw cardiorespiratory tensor $\mathbf{x}_t \in \mathbb{R}^{8\times 750}$, eight channels at 25 Hz. It learns a subject-independent map

$$f_\theta : \left(\mathbf{f}_{1:T}, \mathbf{x}_{1:T}\right) \longmapsto \left(\hat{\mathbf{y}}^{\mathrm{stg}}_{1:T},\ \hat{a}_{1:T}\right), \tag{1}$$

where $\hat{\mathbf{y}}^{\mathrm{stg}}_t$ is a distribution over the five stages and $\hat{a}_t \in [0, 1]$ scores the onset of a respiratory event in epoch t. The AHI estimate is derived from $\hat{a}_{1:T}$ and the decoded stages (Section IV-G). Fig. 2 shows the architecture.

TABLE 2. Stage distribution and respiratory-event prevalence on iSLEEPS ($N = 99$). Respiratory events occupy 15.7% of epochs; 79 of 99 patients exceed 5%.

Stage	W	N1	N2	N3	R
Share (%)	26.7	10.1	42.3	8.9	12.1

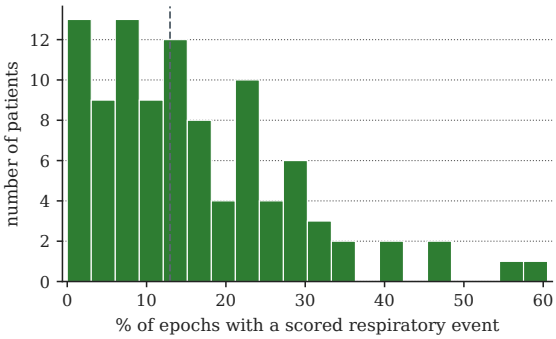


FIGURE 1. Sleep-disordered-breathing burden on iSLEEPS: patients by the fraction of epochs carrying a scored respiratory event. The dashed line marks the median, 13%.

B. NEURAL DESCRIPTORS

Per channel we take absolute and relative Welch band power in delta (0.5–4), theta (4–8), alpha (8–13), sigma (11–16) and beta (16–30) Hz, spectral entropy, the 95% spectral-edge and mean frequency, time-domain statistics and the three Hjorth descriptors [45]; sigma-envelope spindle descriptors and slow-wave amplitude on the EEG, ocular-movement energy on the EOG and tonic level on the EMG complete the 188 (Appendix A). Descriptors are standardized per recording.

C. FINE-TUNED FOUNDATION ENCODER

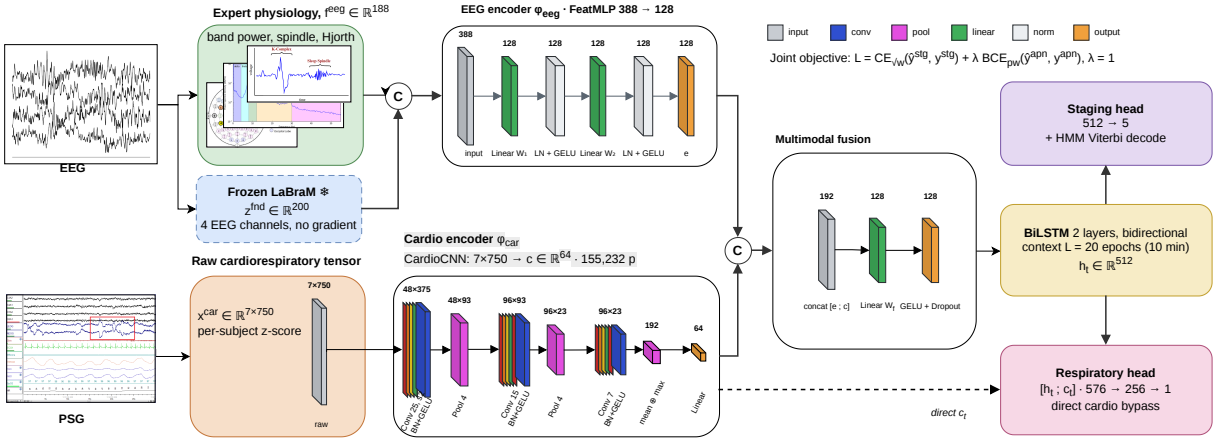


FIGURE 2. MM-Net. The neural stream combines 188 engineered descriptors with the embedding of a LaBraM encoder fine-tuned on EEG and EOG; the cardiorespiratory stream is a convolutional encoder over eight raw channels. The streams are fused by cross-attention and read by a bidirectional LSTM over a 20-epoch window, which feeds a staging head and a respiratory head; the respiratory head also reads the cardiorespiratory embedding directly. Staging is decoded by a transition-regularised Viterbi pass and the AHI is computed from the respiratory scores over predicted sleep.

a: Encoder and input

LaBraM-Base [8] (5.8 M parameters) is pretrained at 200 Hz on 3000-sample windows. Each 30-second epoch is resampled to 200 Hz, split into two 15-second windows, encoded and mean-pooled to $\mathbf{e}_t \in \mathbb{R}^{200}$; amplitudes are scaled to $\mu\mathrm{V}/100$. The four EEG derivations enter as C4, C3, O2 and O1, and the two EOG derivations as F7 and F8, the frontal positions nearest the eyes, so the encoder sees the ocular activity that separates REM from Wake.

b: Fine-tuning

Within each training fold the encoder is fine-tuned with a linear staging head on single epochs of the training patients, with AdamW at 3×10^{-5} for the encoder and 10^{-3} for the head, a one-cycle schedule over three passes, batch size 64 and mixed precision, keeping the state with the best accuracy on the fold's validation patients. Test patients never reach the encoder.

c: Cross-fitting

An encoder fine-tuned on a patient embeds that patient in-sample. If the sequence model is trained on such embeddings and tested on out-of-sample ones, the two distributions differ and the sequence model learns to trust a representation that is cleaner than it will see at test time. We therefore split each fold's training patients into $K = 3$ parts P_1, P_2, P_3 and fine-tune encoder ϕ_k on the training patients outside P_k; every training patient $i \in P_k$ is embedded by ϕ_k, an encoder that never saw it. The full-fold encoder ϕ, fine-tuned on all training patients, embeds the validation and test patients.

d: Alignment

Cross-fitting makes every embedding out of sample but places the training patients in the spaces of K different encoders, which do not coincide with the space of ϕ used at test time: fine-tuning runs from the same initialization, but the resulting coordinates drift. We map each ϕ_k into the space of ϕ with a ridge regression [46] fitted on the patients of P_k, for whom both embeddings exist,

$$W_k = \arg\min_W \left\| (E^{\phi}_{P_k} - \mu^{\phi}_k) - (E^{\phi_k}_{P_k} - \mu^{\phi_k}_k) W \right\|^2_F + \lambda n_k \|W\|^2_F, \tag{2}$$

and use $\tilde{\mathbf{e}}_t = (\mathbf{e}^{\phi_k}_t - \mu^{\phi_k}_k) W_k + \mu^{\phi}_k$ for the training patients, with $\lambda = 0.01$ and n_k the number of epochs in P_k. The map uses only training patients. Every training embedding is thereby out of sample and expressed in the coordinates the sequence model sees at test time. Table 6 shows that both properties are required.

D. CARDIORESPIRATORY ENCODER

A three-layer convolutional encoder reads the raw 8×750 tensor with kernels of 25 samples (one second) and channel widths 48, 96 and 96, each followed by batch normalization and GELU, with max-pooling by four after the first two layers; mean and max pooling over time are concatenated and projected to $c_t \in \mathbb{R}^{64}$. Channels are standardized per recording.

E. FUSION, TEMPORAL DECODER AND HEADS

The neural vector is projected by a two-layer perceptron with LayerNorm to $n_t \in \mathbb{R}^{128}$. The two embeddings are treated as tokens with learned modality embeddings and fused by four-head cross-attention, so each modality reads the other, then projected to $z_t \in \mathbb{R}^{128}$. A two-layer bidirectional LSTM with 256 units per direction reads z_t over a window of $L = 20$ epochs (10 minutes), giving $h_t \in \mathbb{R}^{512}$. The staging head is $\hat{\mathbf{y}}^{\mathrm{stg}}_t = \mathrm{softmax}(W_s h_t)$. The respiratory head reads the temporal state together with a direct copy of the cardiorespiratory embedding,

$$\hat{a}_t = \sigma\big(W_2\,\mathrm{GELU}(W_1[h_t; c_t])\big), \tag{3}$$

so that the mechanical signature of an obstructed breath reaches the respiratory decision without passing through a fu-

TABLE 3. Training configuration, identical across folds and cohorts.

Setting	Value
Foundation encoder (per fold)	
Model	LaBraM-Base, EEG (4) + EOG (2)
Optimizer, LR	AdamW, 3×10^{-5} (head 10^{-3})
Schedule, passes	one-cycle, 3 passes over training epochs
Batch, weight decay	64, 0.05
Cross-fitting, alignment	$K = 3$ parts, ridge $\lambda = 0.01$
MM-Net	
Optimizer, LR, weight decay	AdamW, 3×10^{-4}, 10^{-4}
Schedule	cosine annealing
Window L, batch	20 epochs, 32
Max epochs / patience	45 / 8
Fusion	cross-attention, 4 heads
Staging / respiratory weights	uniform / n_-/n_+
Decoding	Viterbi, training-fold transitions
Seeds	42, 1, 7
Hardware	NVIDIA RTX 3060 Ti (8 GB), PyTorch

sion trained mostly for staging. MM-Net has 2.85 M trainable parameters besides the encoder.

F. TRAINING AND DECODING

Both tasks are trained jointly, $\mathcal{L} = \mathcal{L}^{\mathrm{stg}} + \mathcal{L}^{\mathrm{apn}}$, with an unweighted staging cross-entropy and a respiratory cross-entropy whose positive term is weighted by n_-/n_+. Optimization uses AdamW (3×10^{-4}, weight decay 10^{-4}) with cosine annealing, batch size 32, gradient clipping at 5 and early stopping on the validation patients. Staging posteriors are decoded by a Viterbi pass whose transition matrix and prior are counted from the training labels [47]. Three seeds (42, 1, 7) are trained per fold; we report the single-model mean and the three-seed average of the softmax outputs. Table 3 collects the settings.

G. AHI ESTIMATION

Because $\hat{a}_t$ scores event onsets, its sum over the epochs decoded as sleep estimates the number of events, and dividing by the decoded sleep time gives a rate,

$$\widehat{\mathrm{AHI}} = k \cdot \frac{\sum_{t:\hat{s}_t \neq \mathrm{W}} \hat{a}_t}{\frac{30}{3600}\,|\{t : \hat{s}_t \neq \mathrm{W}\}|}. \tag{4}$$

The positive weighting of the respiratory loss inflates $\hat{a}_t$, so a single scale k is fitted in each fold as the median ratio of annotated to raw estimate over the fold's validation patients and applied unchanged to its test patients. For transfer, k is the median of the ten iSLEEPS fold values (0.459), and nothing is refitted on the external cohort. Severity follows the AASM bands (< 5, 5–15, 15–30, ≥ 30 events/h).

V. EXPERIMENTAL PROTOCOL

On iSLEEPS all results use the same ten patient-independent folds. Within each fold a disjoint set of training patients is held out for validation; validation patients drive every choice (early stopping, encoder state, configuration, AHI scale), and test patients are only scored. ISRUC-I uses ten subject-independent folds of 80 training, 10 validation and 10 test subjects; ISRUC-II and III together use six folds that keep both nights of a subject on the same side; Sleep-EDF uses ten subject-independent folds. In transfer, one model per seed is trained on all 99 iSLEEPS patients (11 for validation) and applied to the external cohort with no re-fitted parameter, threshold or transition matrix.

Staging is scored by accuracy, macro-F1 and Cohen's κ, respiratory detection by the area under the ROC curve (AUC) and average precision (AP), and the AHI by Spearman's ρ, the mean absolute error (MAE), severity-band accuracy and linearly weighted κ. Values are means over the ten test folds with their standard deviation. Paired comparisons use the Wilcoxon signed-rank test on the ten fold means with Holm correction [48] across baselines; with ten folds the smallest attainable two-sided p is 0.002, and 0.016 after Holm correction over eight comparisons.

TABLE 4. Reimplementation check on Sleep-EDF Cassette (ten-fold subject-independent, 40 recordings, Fpz-Cz). Each published figure follows its own paper's protocol; [§] U-Time reports macro-F1 only.

Model	Published	Ours	Δ
TinySleepNet [4]	0.854	0.824 ± 0.045	-0.030
SleepTransformer [21]	0.814	0.823 ± 0.033	$+0.009$
1D-ResNet-SE-LSTM [49]	0.864	0.832 ± 0.036	-0.032
Micro SleepNet [50]	0.828	0.800 ± 0.041	-0.028
AT-BiLSTM [51]	0.838	0.835 ± 0.046	-0.003
U-Time [7][§]	0.79	0.660 ± 0.054	-0.130

a: Baselines

Eight deep architectures are retrained on the iSLEEPS folds: AttnSleep, the corpus LSTM, 1D-ResNet-SE-LSTM, AT-BiLSTM, SleepTransformer, TinySleepNet, U-Time and Huttunen *et al.*, the closest joint staging-and-respiratory system, ported from its authors' code. Each reimplementation reproduces its published Sleep-EDF accuracy (Table 4). For respiratory detection, a desaturation rule, logistic regression and gradient boosting read the 14 engineered cardiorespiratory descriptors on the same folds.

VI. RESULTS ON STROKE SLEEP

A. STAGING AGAINST EVERY BASELINE

Table 5 sets MM-Net beside every architecture we evaluated on iSLEEPS. MM-Net reaches 0.756 ± 0.013 accuracy as a single model and 0.763 ± 0.014 as a three-seed average, with κ 0.656. Every deep architecture developed on healthy sleep stays between 0.599 and 0.720 on this cohort; those we also ran on Sleep-EDF lose 12 to 15 points between healthy and stroke sleep, and the three models trained on healthy sleep and applied directly fall to about 0.31. Paired on the ten fold means, MM-Net is ahead of each of the eight retrained baselines in ten of ten folds, by $+0.042$ (corpus LSTM) to $+0.164$ (Huttunen *et al.*), Holm $p = 0.016$ in every comparison, the minimum this design can return; as a single model the margins are $+0.036$ to $+0.157$, nine or ten folds of ten, Holm $p = 0.016$ in each case. MM-Net is also the only

TABLE 5. Sleep staging on iSLEEPS, patient-independent. Cells are mean ± SD across the ten test folds. † reimplemented and checked on Sleep-EDF (Table 4); ‡ quoted from the corpus paper on its own protocol; * ported from its authors' code; § run on five folds. The Healthy column is ten-fold Sleep-EDF accuracy. MM-Net parameter counts exclude the 5.8 M-parameter encoder. Best in **bold**.

			Healthy	iSLEEPS (stroke)		
Model	Input	Params	Acc	Acc ↑	Macro-F1 ↑	κ ↑
Deep networks, retrained on iSLEEPS						
LSTM (corpus)†	1 EEG + EOG	1.16 M	—	0.720 ± 0.032	0.626 ± 0.028	0.599 ± 0.043
CNN-ResNet18 [20]‡	1 EEG + EOG	∼11 M	—	0.617	0.544	0.480
DeepSleepNet [2]§	1 EEG	∼21 M	—	0.615 ± 0.026	0.567 ± 0.031	0.483 ± 0.038
U-Time [7]†	1 EEG	1.07 M	0.791	0.646 ± 0.064	0.524 ± 0.057	0.506 ± 0.082
AttnSleep [5]	1 EEG	∼0.5 M	—	0.716 ± 0.042	0.600 ± 0.043	0.593 ± 0.058
TinySleepNet [4]†	1 EEG	1.45 M	0.824	0.676 ± 0.050	0.587 ± 0.034	0.548 ± 0.062
SleepTransformer [21]†	1 EEG	2.79 M	0.823	0.683 ± 0.026	0.602 ± 0.030	0.555 ± 0.032
1D-ResNet-SE-LSTM [49]†	1 EEG	0.65 M	0.832	0.708 ± 0.038	0.631 ± 0.033	0.592 ± 0.050
Micro SleepNet [50]†	1 EEG	0.03 M	0.800	0.647 ± 0.049	0.558 ± 0.028	0.510 ± 0.055
AT-BiLSTM [51]†	1 EEG	0.92 M	0.835	0.692 ± 0.040	0.612 ± 0.030	0.570 ± 0.051
CNN + BiLSTM§	4 EEG	∼0.6 M	—	0.613 ± 0.040	0.565 ± 0.034	0.485 ± 0.048
Sleep-EDF transfer§	4 EEG	∼0.6 M	—	0.623 ± 0.020	0.582	0.488
Raw multimodal CNN§	14 ch	0.95 M	—	0.654 ± 0.033	0.610 ± 0.022	0.544 ± 0.037
Huttunen et al. [35]*	4 ch	3.80 M	—	0.599 ± 0.060	0.495 ± 0.067	0.440 ± 0.078
Trained on healthy sleep, applied to iSLEEPS						
CNN	4 EEG	0.49 M	0.806	0.307 ± 0.016	0.223 ± 0.013	0.102 ± 0.010
CNN + BiLSTM (CRNN)	4 EEG	1.15 M	0.828	0.312 ± 0.012	0.253 ± 0.006	0.127 ± 0.022
StagingSeqNet	4 EEG	1.08 M	0.855	0.306 ± 0.009	0.230 ± 0.014	0.091 ± 0.015
Fine-tuned foundation encoder alone (one epoch, no temporal model)						
LaBraM, frozen + logistic regression	4 EEG	5.8 M	—	0.647	—	—
LaBraM, fine-tuned	4 EEG + 2 EOG	5.8 M	—	0.732	—	—
MM-Net						
MM-Net, single model	7 neural + 8 cardioresp.	2.85 M	0.864	0.756 ± 0.013	0.661 ± 0.019	0.647 ± 0.015
MM-Net, three-seed	7 neural + 8 cardioresp.	2.85 M	**0.864**	**0.763 ± 0.014**	**0.664 ± 0.020**	**0.656 ± 0.016**

model in the table that delivers respiratory events and the AHI alongside the stage.

B. WHERE THE GAIN COMES FROM

Table 6 builds MM-Net from its frozen-encoder starting point one change at a time, every row on the same folds and seeds. Uniform class weights and cross-attention fusion contribute +0.009 together. The decisive step is the encoder. A fine-tuned encoder whose training embeddings are cross-fitted but not aligned moves the three-seed accuracy by only +0.002 ($p = 0.63$) and is below the frozen encoder in five folds of ten, by up to 0.023 in the fold where the cross-fitted and full encoders agree least (embedding similarity 0.53). Aligning the spaces converts the same encoder into +0.014 (0.755 → 0.769, $p = 0.020$), and in the single model +0.015 in nine folds of ten ($p = 0.027$). Adding the ocular channels to the encoder raises the encoder's own accuracy from 0.709 to 0.732 and gave the joint-highest validation accuracy (0.779), so it was carried forward; we report the variant validation selected, not the one with the highest test score. Nasal pressure and the epoch alignment of Section III leave staging at 0.763 and serve the respiratory head.

MM-Net is not the encoder alone. Used on its own, the fine-tuned encoder stages single epochs at 0.732; inside MM-Net, with the engineered descriptors, the cardiorespiratory stream and the temporal decoder, the same encoder yields 0.756 as a single model, +0.024 in eight folds of ten ($p = 0.014$). Fig. 3 shows what fine-tuning changes in the representation: held-out epochs that the frozen encoder spreads across one mixed cloud form stage-specific clusters after fine-tuning, with REM, N3 and Wake separated (silhouette by stage −0.045 → 0.047).

TABLE 6. From the frozen-encoder starting point to MM-Net on iSLEEPS, one change per row; same ten folds and three seeds throughout. *p*: Wilcoxon on fold means against the frozen-encoder row with the same fusion.

Configuration	Single	3-seed	*p*
Frozen encoder, class weights, concatenation	0.738	0.749	
+ uniform class weights	0.744	0.754	
+ cross-attention fusion	0.747	0.755	ref.
+ fine-tuned encoder, cross-fitted, not aligned	0.752	0.757	0.63
+ ridge alignment	**0.762**	**0.769**	**0.020**
+ EOG in the encoder (validation-selected)	0.757	0.763	
+ nasal pressure	0.757	0.760	
+ epoch alignment of cardioresp. signals (MM-Net)	0.756	0.763	

C. PER-STAGE PERFORMANCE

Table 7 reports F1 per stage. MM-Net recovers Wake, N2 and REM at 0.79 to 0.82 and N3 at 0.72; REM, the stage the ocular channels in the encoder target, reaches 0.794. N1, the transient stage between wake and sleep, is the hardest stage on this cohort for every method.

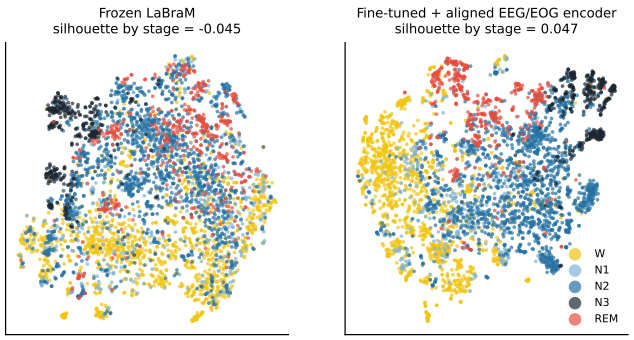


FIGURE 3. t-SNE [52] of 4000 epochs from the ten held-out patients of one fold, coloured by scored stage. Left: frozen LaBraM. Right: the fine-tuned, aligned EEG/EOG encoder used in MM-Net.

TABLE 7. Per-stage F1 on iSLEEPS (three-seed, pooled test epochs).

Model	W	N1	N2	N3	REM
MM-Net	0.810	0.202	0.817	0.716	0.794

TABLE 8. Respiratory-event detection on iSLEEPS, ten-fold patient-independent, same epoch labels for every method. Chance AP is the 0.157 prevalence.

Method	AUC ↑	AP ↑
Desaturation rule	0.595 ± 0.043	0.207
Logistic regression (14 descriptors)	0.603 ± 0.043	0.211
Gradient boosting (14 descriptors)	0.673 ± 0.058	0.290
MM-Net, single model	0.815 ± 0.035	0.471
MM-Net, three-seed	$\mathbf{0.825 \pm 0.035}$	**0.487**

D. RESPIRATORY-EVENT DETECTION

MM-Net detects respiratory-event onsets at 0.825 ± 0.035 AUC and 0.487 ± 0.093 AP (three-seed), against a prevalence of 0.157 (Table 8). On the same folds and labels, gradient boosting on the engineered cardiorespiratory descriptors reaches 0.673, so MM-Net adds 0.152 AUC and 0.197 AP to the strongest conventional detector.

E. AHI ESTIMATION

MM-Net's AHI estimate tracks the annotated index across the full severity range, Spearman $\rho = 0.765$ over the 99 test patients, MAE 10.4 events/h, and places 54% of patients in the correct AASM severity band (weighted κ 0.45; Table 9, Fig. 4). Two comparisons put this in context. The clinical AHI recorded with the corpus agrees with the annotated index at only $\rho = 0.687$, so MM-Net's estimate is closer to the annotations than the clinical index is. Against that clinical index MM-Net reaches $\rho = 0.633$. We also trained a dedicated event detector, a one-dimensional U-Net with a recurrent bottleneck operating at one second on the raw cardiorespiratory signals. Averaged with MM-Net it lowers the MAE to 9.7 and raises severity agreement to 55% (weighted κ 0.51), while MM-Net alone gives the strongest correlation.

TABLE 9. AHI estimation on iSLEEPS, 99 test patients, stages decoded by MM-Net. Severity: AASM bands; κ_w linearly weighted. Last column: ρ against the clinical AHI recorded with the corpus. Reference: agreement of the clinical AHI with the annotated AHI.

Estimator	$\rho \uparrow$	MAE ↓	Sev. acc	κ_w	ρ_{clin}
Event detector	0.669	11.7	0.45	0.44	0.569
MM-Net	**0.765**	10.4	0.54	0.45	**0.633**
MM-Net + event detector	0.742	**9.7**	**0.55**	**0.51**	0.625
Clinical vs. annotated AHI	0.687				

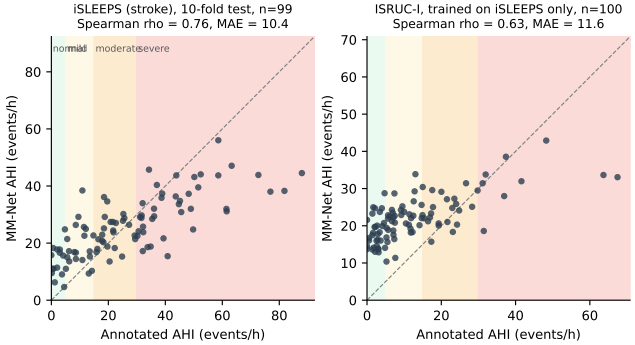


FIGURE 4. MM-Net AHI against the annotated AHI. Left: iSLEEPS, ten-fold test patients. Right: ISRUC-I, MM-Net trained on iSLEEPS only and applied unchanged. Shading marks the AASM severity bands; the dashed line is identity.

F. PHYSIOLOGICAL ATTRIBUTION

We attribute each output to the physiology behind it by intervention, removing one modality at a time and retraining (Table 10; this analysis was run on the frozen-encoder configuration, whose two streams are the same). Removing the cardiorespiratory stream lowers respiratory AUC by 0.128 (Holm $p = 0.018$) and leaves staging untouched; removing the EEG lowers staging by 0.059 (Holm $p = 0.018$) and leaves respiratory detection untouched. Within the cardiorespiratory stream respiratory effort is the one individually necessary channel (-0.038 AUC), the physiological signature of an obstructive event while it happens. Removing the airflow channel the events were scored from leaves detection at 0.778, so the read-out is not a reproduction of the scoring signal. Training both tasks jointly is better than training either alone, by 0.006 staging accuracy ($p = 0.014$) and 0.007 respiratory AUC ($p = 0.049$), with the same architecture and only the loss changed.

VII. GENERALIZATION BEYOND THE DEVELOPMENT COHORT

A. TRAINED AND TESTED ON ISRUC

On ISRUC-I MM-Net, with its encoder fine-tuned, cross-fitted and aligned inside each fold exactly as on iSLEEPS, reaches 0.803 accuracy, macro-F1 0.765 and κ 0.744 (Table 11). It is ahead of the frozen-encoder configuration in nine folds of ten, $+0.018$ ($p = 0.004$), and the gain extends to

TABLE 10. Leave-one-modality-out ablation (frozen-encoder configuration), ten folds, three seeds; *p* Wilcoxon on fold means, Holm-corrected across nine ablations per metric.

Removed	Staging acc	Holm p	Resp. AUC	Holm p
none	0.739	—	0.782	—
EEG	0.680	**0.018**	0.774	0.92
EOG	0.731	0.08	0.777	1.00
EMG	0.735	1.00	0.778	1.00
SpO_2	0.742	1.00	0.780	1.00
respiratory effort	0.741	1.00	0.744	**0.018**
pulse / HRV	0.741	1.00	0.785	1.00
ECG	0.737	1.00	0.777	1.00
airflow	0.740	1.00	0.778	1.00
all cardiorespiratory	0.741	1.00	0.654	**0.018**

respiratory detection (0.872 → 0.889 AUC) and the AHI (ρ 0.523 → 0.590, MAE 6.6 events/h). As on iSLEEPS, MM-Net is ahead of its own encoder used alone (0.792; +0.012, eight folds of ten, $p = 0.014$). The reimplemented baselines, rerun on MM-Net's ISRUC-I folds with three seeds, are listed in the same table. On the smaller ISRUC-II and III, MM-Net reaches 0.725 over 26 nights and 0.779 on the ten healthy subjects of ISRUC-III.

B. HEALTHY SLEEP

On Sleep-EDF MM-Net reaches 0.864 accuracy (macro-F1 0.791, κ 0.810; ten subject-independent folds, three seeds), above every single-channel architecture we reimplemented on the same recordings (0.791–0.835, Table 5). Sleep-EDF carries no cardiorespiratory channels, so this result exercises the neural stream alone. The same architecture therefore leads on healthy sleep, on a sleep-disorder cohort and on stroke sleep.

C. APPLIED WITHOUT RETRAINING

Table 12 applies MM-Net, trained only on iSLEEPS, to 151 external nights from 143 subjects with no re-fitted parameter. On the 100 ISRUC-I patients it stages at 0.730 and detects respiratory events at 0.833 AUC, and its AHI estimate reaches $\rho = 0.631$ (Fig. 4). On UCD, an apnea cohort recorded on a different system with only central EEG, respiratory detection reaches 0.841 AUC and AHI $\rho = 0.653$, with 60% of patients in the correct severity band (weighted κ 0.51); staging there (0.607) reflects the missing occipital and unitless EEG, not the respiratory stream. Respiratory detection transfers essentially intact: 0.83–0.84 AUC on the two apnea cohorts against 0.825 in-cohort.

One deployment observation follows. Nasal pressure is cohort-specific: the eight-channel model, which reads it, transfers with 0.794 respiratory AUC and AHI ρ 0.449 on ISRUC-I, while the seven-channel variant without it reaches 0.833 and 0.631. MM-Net therefore uses all eight channels when trained on a cohort and the seven-channel variant when applied to a new acquisition system.

VIII. DISCUSSION

MM-Net turns the polysomnogram of a stroke patient into the three outputs of a sleep report, hypnogram, respiratory events and AHI, from a single model, and it does so more accurately than any staging architecture we could bring to this cohort. Three findings carry beyond the model.

a: Foundation encoders need out-of-sample, aligned embeddings downstream

The most transferable result is methodological. A fine-tuned foundation encoder is a better stager by itself, yet plugging it into a downstream model gave no significant gain until its training embeddings were cross-fitted and aligned. The failure mode, a downstream model trained on in-sample or misaligned representations, applies to any pipeline that fine-tunes a foundation model and then trains a second model on its outputs, and the remedy costs only K additional fine-tuning runs and a ridge regression per fold. It held on two independent cohorts: +0.014 on iSLEEPS with everything else fixed, and +0.018 for the complete model over the frozen-encoder configuration on ISRUC-I.

b: Stroke sleep is not out of reach

Earlier work on iSLEEPS found every architecture in a narrow band below 0.75 and read it as a property of the injured brain [20]. MM-Net moves past that band on the same patients and folds, 0.763, and reaches 0.803 on a sleep-disorder cohort and 0.864 on healthy sleep with one unchanged architecture. The gap to healthy sleep is narrower than previously reported, and the components that close it, foundation representations adapted to sleep and the full multimodal recording, are identified.

c: The AHI can come from the staging model

A dedicated detector is not needed to obtain a usable AHI. MM-Net's respiratory head, read over its own decoded sleep, estimates the AHI with $\rho = 0.765$ in-cohort and 0.63–0.65 on external cohorts, and its severity bands agree with the annotations at weighted κ up to 0.51. For stroke units without polysomnography technicians, a single automated pass delivering stages, events and severity is the practical form of screening.

d: Scope

The evidence comes from offline whole-night analysis: staging is decoded after the recording ends, which suits overnight polysomnography and not real-time monitoring. Stroke-specific claims rest on iSLEEPS, the only public stroke polysomnography cohort; the four external cohorts establish generalization across populations and acquisition systems. N1 remains the hardest stage on every cohort and for every method, and the AHI severity band is the output with most room to grow; event-level training on larger annotated cohorts is the natural next step for both.

TABLE 11. Trained and tested on ISRUC. ISRUC-I: ten subject-independent folds (80/10/10 subjects), three seeds; baselines rerun on the same folds with three seeds. ISRUC-II+III: six folds keeping both nights of a subject together. Encoder alone: fine-tuned EEG/EOG LaBraM, one epoch, no temporal model. Frozen-encoder configuration: class weights, concatenation, seven cardiorespiratory channels. ISRUC-II+III AHI spans only 0–48 events/h over 26 nights.

Model	Acc ↑	Macro-F1 ↑	κ ↑	Resp. AUC ↑	Resp. AP ↑	AHI ρ ↑	AHI MAE ↓
ISRUC-I (100 *patients*)							
AttnSleep [5]	TBD	TBD	TBD	—	—	—	—
LSTM (corpus)	TBD	TBD	TBD	—	—	—	—
1D-ResNet-SE-LSTM [49]	TBD	TBD	TBD	—	—	—	—
AT-BiLSTM [51]	TBD	TBD	TBD	—	—	—	—
SleepTransformer [21]	TBD	TBD	TBD	—	—	—	—
TinySleepNet [4]	TBD	TBD	TBD	—	—	—	—
U-Time [7]	TBD	TBD	TBD	—	—	—	—
Huttunen et al. [35]	TBD	TBD	TBD	TBD	TBD	—	—
Encoder alone	0.792	—	—	—	—	—	—
Frozen-encoder configuration	0.786	0.752	0.722	0.872	0.447	0.523	6.8
MM-Net	**0.803**	**0.765**	**0.744**	**0.889**	**0.477**	**0.590**	**6.6**
ISRUC-II+III (18 *subjects,* 26 *nights*)							
Frozen-encoder configuration	0.716	0.689	0.632	0.757	0.138	0.061	8.0
MM-Net	**0.725**	**0.692**	**0.641**	**0.794**	**0.197**	**0.138**	**7.3**
of which ISRUC-III (healthy)	0.779	0.747	0.714	0.867	0.260	0.079	3.9

TABLE 12. MM-Net trained on all 99 iSLEEPS patients and applied unchanged (three-seed). Seven-channel variant without nasal pressure; AHI scale k from iSLEEPS. ISRUC-III is healthy, with AHI between 0 and 12.

Cohort	Nights	Acc	κ	Resp. AUC	AHI ρ
ISRUC-I (sleep disorders)	100	0.730	0.641	0.833	0.631
ISRUC-II (sleep disorders)	16	0.681	0.568	0.783	0.518
ISRUC-III (healthy)	10	0.709	0.618	0.776	−0.01
UCD (suspected apnea)	25	0.607	0.463	0.841	0.653
iSLEEPS (in-cohort, reference)	99	0.763	0.656	0.825	0.765

IX. CONCLUSION

MM-Net reads the full polysomnogram and delivers sleep stages, respiratory events and the AHI in one pass. Its fine-tuned EEG foundation encoder, integrated through cross-fitting and ridge alignment, lifts staging on ischemic-stroke sleep to 0.763 accuracy, ahead of every baseline in every fold, and to 0.803 on ISRUC-I and 0.864 on healthy sleep. Respiratory events are detected at 0.825 AUC and the AHI is estimated with $\rho = 0.765$, and both transfer to external cohorts without retraining. Code, features and trained models are released.

APPENDIX A THE 188 NEURAL DESCRIPTORS

Each 30-second epoch is 3000 samples at 100 Hz, and every descriptor below is computed on one epoch alone; the cross-epoch context is supplied by the recurrent decoder. Twenty-three descriptors are computed on each of the seven neural channels, C4:M1, C3:M2, O2:M1, O1:M2, E1:M2, E2:M2 and the chin EMG, which gives 161; a further 27 event descriptors are applied only to the channels whose physiology defines them, giving 188.

The per-channel set begins with the spectrum, a Welch estimate over 256-sample segments overlapping by 128. From it we take absolute and relative power in the five bands of Section IV-B, the logarithm of total power, spectral entropy on the normalised spectrum, the 95% spectral-edge frequency and the mean frequency. Six time-domain statistics follow, namely standard deviation, peak-to-peak, root mean square, skewness, kurtosis and zero-crossing rate, and to these are added the three Hjorth descriptors, activity, mobility and complexity, computed from the first and second differences [45].

The event descriptors follow the graphoelements a scorer reads, and every band-pass among them is a fourth-order Butterworth applied forwards and backwards so that phase is preserved. Spindles contribute three descriptors per EEG channel: the channel is filtered to 11–16 Hz, its analytic envelope is taken by Hilbert transform, and the threshold is set at the 90th percentile of that envelope within the epoch; density counts upward crossings of that threshold, and the mean and standard deviation of the envelope complete the three. Slow waves contribute two per EEG channel, the peak-to-peak and the mean absolute amplitude of the 0.5–4 Hz band. Each EOG channel contributes the log mean squared first difference and the 95th percentile of its absolute value, which separate the fast deflections of REM from slow drift, and the chin EMG contributes log root-mean-square, the 90th percentile of the absolute signal, and the log variance of the first difference, which together track tonic level and its loss in REM. Non-finite values, which arise only from a flat channel, are set to zero, and each descriptor is standardised against the mean and standard deviation of its own recording.

DATA AND CODE AVAILABILITY

iSLEEPS is publicly available (Zenodo, DOI 10.5281/zenodo.14873844) [19]; ISRUC-Sleep [41], the UCD database [42] and Sleep-EDF [44] are public. Our code, the per-fold configuration, engineered features and the corrections applied to the external cohorts are released at https://github.com/wshuv-o/isleeps-sleep-staging.

ETHICS

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th

IEC (BS&NS DIV.)/2022, dated 05.02.2022) [19]. This work is a secondary analysis of de-identified public data and required no additional enrollment.

FUNDING

The authors received no specific funding for this work.

ACKNOWLEDGMENT

The authors thank the iSLEEPS, ISRUC-Sleep, UCD and Sleep-EDF investigators for releasing their data.

REFERENCES

[1] American Academy of Sleep Medicine, *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications, Version 3*. American Academy of Sleep Medicine, 2023.

[2] A. Supratak, H. Dong, C. Wu, and Y. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 1998–2008, 2017. [Online]. Available: https://doi.org/10.1109/TNSRE.2017.2721116

[3] H. Phan, F. Andreotti, N. Cooray, O. Y. Chén, and M. De Vos, "SeqSleepNet: End-to-end hierarchical recurrent neural network for sequence-to-sequence automatic sleep staging," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 27, no. 3, pp. 400–410, 2019. [Online]. Available: https://doi.org/10.1109/TNSRE.2019.2896659

[4] A. Supratak and Y. Guo, "TinySleepNet: An efficient deep learning model for sleep stage scoring based on raw single-channel EEG," in *42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2020, pp. 641–644. [Online]. Available: https://doi.org/10.1109/EMBC44109.2020.9176741

[5] E. Eldele, Z. Chen, C. Liu, M. Wu, C.-K. Kwoh, X. Li, and C. Guan, "An attention-based deep learning approach for sleep stage classification with single-channel EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 809–818, 2021. [Online]. Available: https://doi.org/10.1109/TNSRE.2021.3076234

[6] H. Phan, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, and M. De Vos, "XSleepNet: Multi-view sequential model for automatic sleep staging," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 9, pp. 5903–5915, 2022. [Online]. Available: https://doi.org/10.1109/TPAMI.2021.3070057

[7] M. Perslev, M. H. Jensen, S. Darkner, P. J. Jennum, and C. Igel, "U-Time: A fully convolutional network for time series segmentation applied to sleep staging," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 4417–4428.

[8] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," in *International Conference on Learning Representations (ICLR)*, 2024, arXiv:2405.18765.

[9] J. Wang, S. Zhao, Z. Luo, Y. Zhou, H. Jiang, S. Li, T. Li, and G. Pan, "CBraMod: A criss-cross brain foundation model for EEG decoding," in *International Conference on Learning Representations (ICLR)*, 2025, arXiv:2412.07236.

[10] G. Wang, W. Liu, Y. He, C. Xu, L. Ma, and H. Li, "EEGPT: Pretrained transformer for universal and reliable representation of EEG signals," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024. [Online]. Available: https://doi.org/10.52202/079017-1239

[11] C. Yang, M. B. Westover, and J. Sun, "BIOT: Biosignal transformer for cross-data learning in the wild," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 78 240–78 260. [Online]. Available: https://openreview.net/forum?id=c2LZyTyddi

[12] H. Phan and K. Mikkelsen, "Automatic sleep staging of EEG signals: Recent development, challenges, and future directions," *Physiological Measurement*, vol. 43, no. 4, 2022. [Online]. Available: https://doi.org/10.1088/1361-6579/ac6049

[13] K. Yang and H. Wang, "Deep learning for EEG-based sleep stage classification: A review," *Medical & Biological Engineering & Computing*, vol. 64, pp. 2017–2043, 2026. [Online]. Available: https://doi.org/10.1007/s11517-026-03583-3

[14] C. L. Bassetti and M. S. Aldrich, "Sleep electroencephalogram changes in acute hemispheric stroke," *Sleep Medicine*, vol. 2, no. 3, pp. 185–194, 2001. [Online]. Available: https://doi.org/10.1016/s1389-9457(00)00071-x

[15] C. Denis, I. Jaussent, L. Guiraud, C. Mestejanot, C. Arquizan, I. Mourand, S. Chenini, B. Abril, A. Wacongne, R. Tamisier, S. Baillieul, J.-L. Pepin, L. Barateau, and Y. Dauvilliers, "Functional recovery after ischemic stroke: Impact of different sleep health parameters," *Journal of Sleep Research*, vol. 33, no. 1, 2024. [Online]. Available: https://doi.org/10.1111/jsr.13964

[16] J. M. Gottselig, C. L. Bassetti, and P. Achermann, "Power and coherence of sleep spindle frequency activity following hemispheric stroke," *Brain*, vol. 125, no. 2, pp. 373–383, 2002. [Online]. Available: https://doi.org/10.1093/brain/awf021

[17] L. M. J. Fernández and A. Lüthi, "Sleep spindles: Mechanisms and functions," *Physiological Reviews*, vol. 100, no. 2, pp. 805–868, 2020. [Online]. Available: https://doi.org/10.1152/physrev.00042.2018

[18] L. Facchin, C. Schöne, A. Mensen *et al.*, "Slow waves promote sleep-dependent plasticity and functional recovery after stroke," *Journal of Neuroscience*, vol. 40, no. 45, pp. 8637–8651, 2020. [Online]. Available: https://doi.org/10.1523/jneurosci.0373-20.2020

[19] S. Maiti, S. K. Sharma, S. Mythirayee, S. Rajendran, and R. S. Bapi, "Polysomnography dataset for sleep analysis in ischemic stroke patients," *Scientific Data*, vol. 13, no. 1, p. 421, 2026. [Online]. Available: https://doi.org/10.1038/s41597-026-06747-w

[20] S. Bose, S. Maiti, S. K. Sharma, S. Mythirayee, T. Chakraborti, S. Rajendran, and R. S. Bapi, "AI generalisation gap in comorbid sleep disorder staging," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026. [Online]. Available: https://doi.org/10.1109/ISBI61048.2026.11515484

[21] H. Phan, K. Mikkelsen, O. Y. Chén, P. Koch, A. Mertins, and M. De Vos, "SleepTransformer: Automatic sleep staging with interpretability and uncertainty quantification," *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 8, pp. 2456–2467, 2022. [Online]. Available: https://doi.org/10.1109/TBME.2022.3147187

[22] H. Phan, K. P. Lorenzen, E. Heremans, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, M. Baumert *et al.*, "L-SeqSleepNet: Whole-cycle long sequence modelling for automatic sleep staging," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 10, pp. 4748–4757, 2023. [Online]. Available: https://doi.org/10.1109/JBHI.2023.3303197

[23] N. Jirakittayakorn, Y. Wongsawat, and S. Mitrirattanakul, "ZleepAnlystNet: A novel deep learning model for automatic sleep stage scoring based on single-channel raw EEG data using separating training," *Scientific Reports*, vol. 14, no. 1, p. 9859, 2024. [Online]. Available: https://doi.org/10.1038/s41598-024-60796-y

[24] Y. Guo, M. Nowakowski, and W. Dai, "FlexSleepTransformer: A transformer-based sleep staging model with flexible input channel configurations," *Scientific Reports*, vol. 14, no. 1, p. 26312, 2024. [Online]. Available: https://doi.org/10.1038/s41598-024-76197-0

[25] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv preprint arXiv:2312.00752*, 2023.

[26] X. Zhou, Y. Han, Z. Chen, C. Liu, Y. Ding, Z. Jia, and Y. Liu, "BiT-MamSleep: Bidirectional temporal mamba for EEG sleep staging," *arXiv preprint arXiv:2411.01589*, 2024. [Online]. Available: https://doi.org/10.48550/ARXIV.2411.01589

[27] M. Perslev, S. Darkner, L. Kempfner, M. Nikolic, P. J. Jennum, and C. Igel, "U-Sleep: Resilient high-frequency sleep staging," *npj Digital Medicine*, vol. 4, no. 1, p. 72, 2021.

[28] R. Thapa, M. R. Kjaer, B. He, I. Covert, H. Moore IV, U. Hanif *et al.*, "A multimodal sleep foundation model for disease prediction," *Nature Medicine*, vol. 32, no. 2, pp. 752–762, 2026. [Online]. Available: https://doi.org/10.1038/s41591-025-04133-4

[29] Y. Zhou, J. Wu, Z. Ren, Z. Yao, W. Lu, K. Peng, Q. Zheng, C. Song *et al.*, "CSBrain: A cross-scale spatiotemporal brain foundation model for EEG decoding," *arXiv preprint arXiv:2506.23075*, 2025. [Online]. Available: https://doi.org/10.52202/085713-2915

[30] K. Kontras, T. Osselaer, S. G. Mouslech, A.-I. Karaiskou, G. Gagliardi, T. Strypsteen *et al.*, "NeuroAtlas: Benchmarking foundation models for clinical EEG and brain-computer interfaces," *arXiv preprint arXiv:2605.14698*, 2026.

[31] A. Dei Rossi, M. Metaldi, M. Bechny, I. Filchenko, J. van der Meer, M. H. Schmidt, C. L. A. Bassetti, A. Tzovara *et al.*, "SLEEPYLAND: Trust begins with fair evaluation of automatic sleep staging models,"

npj Digital Medicine, vol. 9, no. 1, p. 55, 2026. [Online]. Available: https://doi.org/10.1038/s41746-025-02237-2

[32] E. Estevan, M. Sierra-Torralba, E. López-Larraz, and L. Montesano, "A systematic evaluation of self-supervised learning for label-efficient sleep staging with wearable EEG," *arXiv preprint arXiv:2510.07960*, 2025. [Online]. Available: https://doi.org/10.48550/ARXIV.2510.07960

[33] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[34] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins, "Double/debiased machine learning for treatment and structural parameters," *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68, 2018.

[35] R. Huttunen, T. Leppänen, B. Duce, E. S. Arnardottir, S. Nikkonen, S. Myllymaa, and J. Töyräs, "A comparison of signal combinations for deep learning-based simultaneous sleep staging and respiratory event detection," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 5, pp. 1704–1714, 2023. [Online]. Available: https://doi.org/10.1109/TBME.2022.3225268

[36] "A multi-task deep learning approach for simultaneous sleep staging and apnea detection for elderly people," *Interdisciplinary Sciences: Computational Life Sciences*, 2025. [Online]. Available: https://doi.org/10.1007/s12539-025-00721-7

[37] N. Tellenbach, M. H. Schmidt, F. Alexiev, E. Blondiaux, F. Cavalloni, C. L. Bassetti, L. Heydrich, and P. Bargiotas, "REM sleep and muscle atonia in brainstem stroke: A quantitative polysomnographic and lesion analysis study," *Journal of Sleep Research*, vol. 32, no. 2, 2023.

[38] B. K. Simpson, R. Rangwani, A. Abbasi, J. M. Chung, C. M. Reed, and T. Gulati, "Disturbed laterality of non-rapid eye movement sleep oscillations in post-stroke human sleep: A pilot study," *Frontiers in Neurology*, vol. 14, p. 1243575, 2023. [Online]. Available: https://doi.org/10.3389/fneur.2023.1243575

[39] Y. Zheng, T. Jing, L. Xu, S. Sun, W. Yu, R. Liu, G. Xu, and C. Chu, "Analysis of the correlation between sleep spindles and cognitive impairment in patients with ischemic stroke," *Scientific Reports*, vol. 15, no. 1, p. 36259, 2025. [Online]. Available: https://doi.org/10.1038/s41598-025-20102-w

[40] A. Agius Anastasi *et al.*, "Brain symmetry index in healthy and stroke patients for assessment and prognosis," *Stroke Research and Treatment*, vol. 2017, 2017. [Online]. Available: https://doi.org/10.1155/2017/8276136

[41] S. Khalighi, T. Sousa, J. M. Santos, and U. Nunes, "ISRUC-Sleep: A comprehensive public dataset for sleep researchers," *Computer Methods and Programs in Biomedicine*, vol. 124, pp. 180–192, 2016.

[42] St Vincent's University Hospital and University College Dublin, "St Vincent's University Hospital / University College Dublin Sleep Apnea Database," PhysioNet, 2007.

[43] A. L. Goldberger, L. A. N. Amaral, L. Glass *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000. [Online]. Available: https://doi.org/10.1161/01.cir.101.23.e215

[44] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Oberyé, "Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the EEG," *IEEE Transactions on Biomedical Engineering*, vol. 47, no. 9, pp. 1185–1194, 2000. [Online]. Available: https://doi.org/10.1109/10.867928

[45] B. Hjorth, "EEG analysis based on time domain properties," *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970. [Online]. Available: https://doi.org/10.1016/0013-4694(70)90143-4

[46] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[47] W. Chen, Y. Cai, A. Li, Y. Su, and K. Jiang, "Single-channel sleep EEG classification method based on LSTM and hidden markov model," *Brain Sciences*, vol. 14, no. 11, p. 1087, 2024. [Online]. Available: https://doi.org/10.3390/brainsci14111087

[48] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[49] W. Li and J. Gao, "Automatic sleep staging by a hybrid model based on deep 1D-ResNet-SE and LSTM with single-channel raw EEG signals," *PeerJ Computer Science*, vol. 9, p. e1561, 2023. [Online]. Available: https://doi.org/10.7717/peerj-cs.1561

[50] G. Liu, G. Wei, S. Sun, D. Mao, J. Zhang, D. Zhao, X. Tian, X. Wang, and N. Chen, "Micro SleepNet: Efficient deep learning model for mobile terminal real-time sleep staging," *Frontiers in Neuroscience*, vol. 17, p. 1218072, 2023. [Online]. Available: https://doi.org/10.3389/fnins.2023.1218072

[51] M. Fu, Y. Wang, Z. Chen, J. Li, F. Xu, X. Liu, and F. Hou, "Deep learning in automatic sleep staging with a single channel electroencephalography," *Frontiers in Physiology*, vol. 12, p. 628502, 2021. [Online]. Available: https://doi.org/10.3389/fphys.2021.628502

[52] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.



MD WAHIDUZZAMAN SUVA received the B.Sc. degree in computer science and engineering from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2025, where he is currently pursuing the M.Sc. degree in computer science. He is an Applied AI Engineer with Odin, Dhaka, and a Co-Founder of Prodigy AI, a non-profit research group focused on brain–computer interfaces. He is a co-author of an article on equivariant quantum graph neural networks in *IEEE Transactions on Artificial Intelligence*, of an article on equivariant quantum clustering in *Physica Scripta*, and of a Springer book chapter on federated learning in healthcare technology. His work spans quantum machine learning, federated learning, and EEG-based brain–computer interfaces. He serves as a reviewer for several international journals and conferences. His research interests include brain–computer and brain–machine interfaces, brain signal analysis, transformer models for EEG, and applied machine learning for health.



ESM-E MOULA CHOWDHURY ABHA received the B.Sc. degree in computer science and engineering from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2025, where she is currently pursuing the M.Sc. degree in computer science. She is with the Department of Computer Science and Engineering, AIUB, and a Researcher with Prodigy AI, a non-profit research group focused on brain–computer interfaces. She has over three years of industry experience as a Backend and AI Application Developer, building scalable API architectures and AI-integrated platforms for fintech, education, and customer-support products. She is the lead author of a Springer book chapter on federated learning in healthcare technology and a co-author of an article on equivariant quantum clustering in *Physica Scripta*, and she has contributed to research on attention-based EEG classification and multimodal deep learning for autonomous vehicle perception. Her research interests span brain signal analysis, transformer models for EEG, brain–computer and brain–machine interfaces, and applied machine learning for health.



MD IMTIAJ ALAM SAJIN received the B.Sc. degree in computer science and engineering, with a major in data science, from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2026, where he is currently pursuing the M.Sc. degree in computer science. He is with the Department of Computer Science and Engineering, AIUB, and a Researcher with Prodigy AI, a non-profit research group focused on brain–computer interfaces. He has also worked as a Software Developer with Byteverse Ltd., Dhaka, contributing to Chatsby, an AI-driven customer-support platform. His B.Sc. thesis proposed a dual-attention transformer for EEG classification in brain–computer interfaces. He is a co-author of an article on symmetry-aware equivariant quantum clustering in *Physica Scripta* and of a Springer book chapter on federated learning in healthcare technology. His research interests are in transformer models for EEG, brain–computer and brain–machine interfaces, brain signal analysis, and applied machine learning for health.



MD IFTEKHARUL MOBIN (Member, IEEE) is an Assistant Professor with the Faculty of Science and Technology, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. He received the Ph.D. degree from Queen Mary University of London, London, U.K., and the B.Sc. degree from the Islamic University of Technology (IUT), Gazipur, Bangladesh. He was awarded the Queen Mary scholarship and the ImpactQM scholarship in acknowledgment of his contributions to industry–academia partnerships, and his research has received support from Bangladesh's Ministry of Information and Communication Technology. He served as a software engineer in the United Kingdom, Japan, and Bangladesh for several years, and that dual exposure to academic and industry settings informs his perspective on industry–academia collaboration. He has served on the program committees of international conferences and as an editor and reviewer for conferences and journals worldwide. His research interests include natural language processing, time-series prediction, health informatics, the Internet of Things, and image processing.